MIL-F-62564(AT)
28 April 1988
SUPERSEDING
See (6.5)

MILITARY SPECIFICATION

FAN, VANEAXIAL, ENGINE COOLING

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a vaneaxial fan which is a component of the engine cooling system in ground combat vehicles. The vane axial fan will be referred to by the term "fan" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4140

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SPECIFICATIONS

FEDERAL

P-C-437	- Cleaning Compound, High Pressure (Steam) Cleaner.
GG-P-455	- Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
QQ-P-35	- Passivation Treatments for Corrosion-Resisting Steel.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
TT-C-490	- Cleaning Methods for Ferrous Surface and Pretreatment for Organic Coatings.

MILITARY

MIL-P-514	- Plates Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-B-17931	- Bearings, Ball, Annular, for Quiet Operation.
MIL-P-19834	- Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for.
MIL-C-22750	- Coating Epoxy – Polyamide.
MIL-P-23377	- Primer Coatings: Epoxy – Polyamide, Chemical and Solvent Resistant.
MIL-P-52192	- Primer Coating, Epoxy.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.
MIL-P-53030	- Primer Coatings, Epoxy, Water Reducible, Lead and Chromate Free.
MIL-G-81322	- Grease, Aircraft, General Purpose Wide Temperature Range, NATO Code Number G-395.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWINGS
ARMY

10929029	- Gage, Involute Go Composite Plug.
10929030	- Gage, Involute Not Go Sector Plug.
12292080	- Fan, Vane Axial.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

AMCA-STD-210	- Laboratory Methods for Testing Fans for Rating Purposes.

(Applications for copies should be addressed to the Air Moving and Conditioning Association, Inc., 30 West University Drive, Arlington Heights, Illinois 60004.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	- Standard Specification for Electrode-Positive Coatings of Zinc on Iron and Steel.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish fans which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent fans delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards and drawings. Materials shall be uniform and free of defects and imperfections that might affect their serviceability or the performance of the finished product (see 4.7.1).

3.2.1 Metals. All metals shall be compatible with the temperature and with the functional, service, and storage conditions to which the components will be exposed (see 4.7.1).

3.2.1.1 Dissimilar metals. The use of dissimilar metals shall be in accordance with MIL-STD-889 (see 4.7.1).

3.2.2 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

3.3 Design and construction. The fan shall be fabricated as specified herein and in accordance with Drawing 12292080 (see 4.7.1 and 4.7.2).

3.3.1 Weight. The weight of the fan shall not exceed 125 pounds (lb) (see 4.7.3).

3.3.2 Balance. The fan impeller shall be dynamically balanced to within 0.031 inch-ounce (see 4.7.1).

3.3.3 Bearings. Bearings shall be in accordance with MIL-B-17931, except lubricant shall conform to MIL-G-81322. Bearings shall be lubricated for life and shall be either integrally sealed or shall have single seals and lubrication cavity within the impeller hub (see 4.7.2).

3.3.4 Protective coatings.

3.3.4.1 Steel components. Except for shaft and internal parts, exposed steel components shall be zinc plated in accordance with ASTM B633 or cadmium plated in accordance with QQ-P-416, type II, class 3 and exposed corrosion resisting steel components shall be passivated in accordance with QQ-P-35 (see 4.7.2).

3.3.5 Spline. Spline hardness shall be as specified on Drawing 12292080 (see 4.7.2).

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3.4 Performance.

3.4.1 Direction of rotation. The fan shall operate clockwise when it is viewed from the output end. The direction of rotation and the direction of airflow shall be permanently marked on the fan housing. There shall be no rubbing or binding when the impeller is rotated slowly (see 4.7.4).

3.4.2 Capacity. The fan shall deliver not less than 22,050 cubic feet per minute of air weighing 0.0579 pounds per cubic foot at a static pressure of 15.0 plus or minus (±) 0.5 inches of water and a total pressure of 18.0 ± 0.5 inches of water with the fan rotation no greater than 5100 revolutions per minute (rpm) (see 4.7.5).

3.4.3 Torque. Drive torque shall not exceed 1,150 inch pounds when the fan is operated at 5075 ± 25 rpm with air output as specified in 3.4.2 (see 4.7.6).

3.4.4 Noise level. The sound power level of the fan when operating at 5100 rpm with air output as specified in 3.4.1, shall not exceed the values shown in table I, when measured at a distance of 10 feet in any direction (see 4.7.7).

TABLE I. Sound power level.

Octave band (Hz)	Sound power level (dB re 10^{-12}W) 1/
63	110
125	108
250	115
500	130
1000	130
2000	128
4000	125

1/ Decibel (dB), reference (re), Watt (W).

3.4.5 Environmental.

3.4.5.1 High temperature. The fan shall demonstrate no performance degradation and show no evidence of damage or deformation after being stored and operated at temperatures up to 225 degrees Fahrenheit (°F) (see 4.7.8.1).

3.4.5.2 Low temperature. The fan shall demonstrate no performance degradation and show no evidence of damage or deformation after being stored at temperatures down to minus (-) 70°F and operated at temperatures down to -65°F (see 4.7.8.2).

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3.4.5.3 Steam and waterjet cleaning. The fan shall show no evidence of damage or deterioration and demonstrate no performance degradation following a steam and waterjet cleaning process which uses a cleaner conforming to P-C-437, and which applies the jet perpendicular to the fan surface from a distance of not more than 1 foot for steam cleaning and not more than 5 feet for waterjet cleaning. The jet pressure shall be 105 ± 5 pound per square inch gage (psig) and the jet shall be moved at such a rate that the entire surface of the fan is subjected to the jet at a rate of 1 square foot per minute (see 4.7.8.3).

3.4.5.4 Shock. The fan shall show no evidence of damage or deformation and demonstrate no performance degradation after exposure to sawtooth shock pulses of 40 gravity units (g) for 11 milliseconds (ms) time duration applied along each of three major axes (see 4.7.8.4).

3.4.5.5 Vibration. The fan shall show no evidence of damage or deformation and demonstrate no performance degradation after exposure to vibration levels of 1.5 and 4.2 g in the sweep frequency range of 5 to 500 to 5 Hz with a sweeptime of 15 minutes (see figure 1 and 4.7.8.5).

3.4.6 Durability.

3.4.6.1 Fan system compatibility. To assure fan compatibility with mating system, performance under vehicle operational environment and adequate durability, the fan shall be installed in the vehicle and operated for a minimum of 6,000 miles. The mileage shall be apportioned over the road conditions specified below. The fan shall meet the requirements of 3.4.1 through 3.4.4 after the durability test (see 4.7.9.1).

Road conditions	Operating miles (minimum)
Paved roads	1,260 miles
Secondary roads	2,700 miles
Cross-country	2,040 miles

3.4.6.2 Durability measurement. The durability of the fan when installed in the vehicle and operated as specified in 3.4.6.1 shall be demonstrated based on a minimum of 6,000 miles of operation without failure requiring replacement of the fan. Failure is defined as the inability of the fan to meet its required functions during operation and subsequently meeting the performance requirements of 3.4.1 through 3.4.4 (see 4.7.9.2).

3.5 Finish. All exposed ferrous and aluminum surfaces shall be cleaned, treated, primed and finished as specified herein (see 4.7.2).

3.5.1 Ferrous metals.

3.5.1.1 Cleaning and treatment.

a. Clean ferrous metals (Rockwell hardness less than C40) in accordance with the applicable method (I through VI) specified in TT-C-490. Treat in accordance with type I or III of TT-C-490.

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b. Clean ferrous metals (Rockwell hardness C40 through C47) in accordance with the applicable method (I through VI) specified in TT-C-490. Treat in accordance with type I or III of TT-C-490 (bake for hydrogen embrittlement relief as required).

c. Clean ferrous metals (Rockwell hardness C48 and over) in accordance with method I of TT-C-490. Acid solutions shall not be used for cleaning.

3.5.1.2 Priming. Apply a primer coating in accordance with MIL-P-53030, MIL-P-53022 or MIL-P-52192, composition optional. The dry film thickness of the primer shall be not less than 1.0 mil.

3.5.1.3 Finishing. Apply an epoxy-polyamide coating conforming to type I of MIL-C-22750 or a modified high solids epoxy which meet the performance requirements of MIL-C-22750. The dry film thickness of the coating shall be not less than 1.0 mil. The color shall match color chip number 24533 (green) of FED-STD-595.

3.5.2 Aluminum.

3.5.2.1 Treatment. Treat in accordance with class 1A of MIL-C-5541.

3.5.2.2 Priming. Apply a primer coating in accordance with MIL-P-53030, MIL-P-53022 or MIL-P-23377, composition optional. The dry film thickness of the primer shall be not less than 1.0 mil.

3.5.2.3 Finishing. Apply an epoxy-polyamide coating conforming to type I of MIL-C-22750 or a modified high solids epoxy which meets the performance requirements of MIL-C-22750. The dry film thickness of the coating shall be not less than 1.0 mil. The color shall match color chip number 24533 (green) of FED-STD-595.

3.6 Identification marking.

3.6.1 Nameplates. Nameplates shall be as specified in either a or b:

a. Material and construction shall conform to MIL-P-514, type III, composition A, class 2, or composition C. Composition C material shall conform to GG-P-455, type II, grade A, class 1. The nameplate shall be attached with suitable metal fasteners to achieve a permanent installation (see 4.7.2).
b. Adhesive-backed aluminum foil shall conform to MIL-P-19834, type I, glossy, except backing sheet, which may be of paper, shall be removable intact by stripping without use of water or other solvents (see 4.7.2).

3.6.2 Product marking. The fan shall be identified in accordance with MIL-STD-130 and Drawing 12292080 (see 4.7.2).

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3.7 Workmanship. Workmanship shall be of the quality necessary to provide fans free from defects that would affect their appearance, performance or present safety hazards to operating and maintenance personnel (see 4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent (%) of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Test (see 4.5.3).

c. Control tests (see 4.6).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 ± 18°F
b. Barometric pressure 28.5 + 2 inches mercury (in Hg)
   – 3
c. Relative humidity 50 ± 30%

4.3.1 Environmental tests.

4.3.1.1 Test temperature stabilization. Test temperature stabilization is the point at which the test sample will not vary more than 3.6°F per hour.

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply fans that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two fans. Preproduction inspection shall consist of inspection as specified in table II. When specified (see 6.2), durability testing by the Government will require two additional preproduction samples that have successfully completed the examination and performance tests (see table II).

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select three fans produced under the production contract for initial production inspection. Initial production units shall be inspected as specified in table II.

4.4.3 First article inspection failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article inspection is completed to the satisfaction of the Government.

4.4.4 Disposition of sample fans. Unless otherwise specified (see 6.2), samples which have been subjected to first article inspection shall not be delivered on the contract.

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4.5 Quality conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the fans of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

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TABLE II. Classification of inspectoin.

Title	Requirement	Inspec-tion	Production Inspection				Initial Production			Quality Conformance		Con-trol
			Sample Units				Sample Units			Exami-nation	Test	
			1	2	3	4	1	2	3			
CONTRACTOR INSPECTION												
Materials and construction	3.2 thru 3.2.1.1, 3.3 and 3.3.2	4.7.1	X	X	X	X	X	X	X			
Defects (see table III)	3.3, 3.3.3, 3.3.4, 3.3.5, and 3.7	4.7.2	X	X	X	X	X	X	X	X		
Weight	3.3.1	4.7.3	X	X			X	X	X			
Direction of rotarion	3.4.1	4.7.4	X	X	X	X	X	X	X		X	
Capacity	3.4.2	4.7.5	X	X	X	X	X	X	X			
Torque	3.4.3	4.7.6	X	X	X	X	X	X	X			
Noise level	3.4.4	4.7.7	X	X	X	X	X	X	X			
High temperature	3.4.5.1	4.7.8.1	X	X			X	X	X			
Low temperature	3.4.5.2	4.7.8.2	X	X			X	X	X			
Steam and water jet cleaning	3.4.5.3	4.7.8.3	X	X			X	X	X			
Shock	3.4.5.4	4.7.8.4	X	X			X	X	X			X
Vibration	3.4.5.5	4.7.8.5	X	X			X	X	X			X
GOVERNMENT INSPECTION												
Durability	3.4.6	4.7.9			X	X						

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4.5.2 Examination.

4.5.2.1 Acceptable quality level (AQL). Each sample selected in accordance with 4.5.2.1 shall be examined to determine conformance to the following AQL's.

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Nonconformance in design construction (see 3.3 and 3.3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
103	Nonconformance of spline minor diameter (see 3.3).	SIE
104	Nonconformance of involute spline (see 3.3)	2/
105	Spline surface hardness (see 3.3.5).	3/
106	Faulty workmanship affecting performance or presenting safety hazards (see 3.7).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE
202	Improper protective coatings (see 3.3.4).	Visual
203	Improper finish (see 3.5).	Visual
204	Improper marking (see 3.6).	Visual
205	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment
2/ Involute go composite plug gage confirming to Drawing 10929029 and involute not go sector plug gage conforming to Drawing 10929030.
3/ Hardness tester.

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4.5.2.3 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective fans and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.3 Test. Each fan shall be subjected to the quality conformance test specified in table II.

4.5.3.1 Failure. Failure of any fan to pass the specified quality conformance test shall be cause for the Government to refuse acceptance of that fan.

4.6 Control tests. Unless otherwise specified (see 6.2), sample fans shall be selected at the rate of 1 per each 100 fans produced. The fans shall be subjected to the control test specified in table II.

4.6.1 Failure. If any fan fails to pass any of the control tests, two additional fans may be selected from the lot and subjected to the control tests. Failure of either fan to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quality represented until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.6.2 Disposition of sample fans. Fans which have been subjected to control testing, shall not be delivered on the contract.

4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.2 through 3.2.1.1, 3.3 and 3.3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3, 3.3.3, 3.3.4, 3.3.5 and 3.5 through 3.7 shall be determined by examination for the defects listed in table III. Examination shall be visual or by measurement with standard inspection equipment.

4.7.3 Weight. To determine conformance to 3.3.1, the fan shall be weighed.

4.7.4 Direction of rotation. To determine conformance to 3.4.1, the impeller shall be rotated slowly and examined for rubbing or binding. Rotate impeller clockwise as specified and verify that airflow is in the direction specified on Drawing 12292080. Verify that direction of rotation and direction of airflow is properly marked on fan housing.

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4.7.5 Capacity. To determine conformance to 3.4.2, the fan's airflow rate and total pressure shall be determined in accordance with AMCA-STD-210.

4.7.6 Torque. To determine conformance to 3.4.3, drive torque shall be measured using a dynamometer while the fan is mounted on a suitable test stand and operating at 5075 ± 25 rpm with air output as specified in 3.4.2.

4.7.7 Noise level. To determine conformance to 3.4.4, the octave band sound power levels shall be measured in accordance with accepted standard practices for air moving devices.

4.7.8 Environmental.

4.7.8.1 High temperature. To determine conformance to 3.4.5.1, the fan shall be installed in a chamber in a manner that will simulate service usage. Raise the internal chamber temperature to 225°F and maintain for a period of 48 hours while insuring the relative humidity is not in excess of 15%. Then, with the internal chamber temperature at 225°F, the highest operating temperature for which the fan is designed to operate, the impeller shall be rotated slowly and examined for rubbing and binding. When raising or lowering the temperature of the chamber, the rate of temperature change shall not exceed 18°F per minute. At the conclusion of the test, the fan shall be returned to standard ambient conditions, operated at 5075 rpm for 15 minutes to verify that the impeller rotates smoothly and examined for evidence of damage or deformation.

4.7.8.2 Low temperature. To determine conformance to 3.4.5.2, the fan shall be installed in a chamber in a manner that will simulate service usage. Lower the internal chamber temperature to -70°F and maintain for a period of 24 hours after temperature stabilization (see 4.3.1.1). Then, raise the internal chamber temperature to -65°F, the lowest temperature for which the fan is designed to operate, and maintain until temperature stabilization of the fan is reached. The impeller shall be rotated slowly and examined for rubbing and binding. When raising or lowering the temperature in the chamber, the rate of temperature change shall not exceed 18°F per minute. At the conclusion of the test, the fan shall be returned to standard ambient conditions, operated at 5075 rpm for 15 minutes to verify that the impeller rotates smoothly and examined for evidence of damage or deformation.

4.7.8.3 Steam and waterjet cleaning. To determine conformance to 3.4.8.3, the fan shall be subjected to a cleaning as specified therein. Within 1 hour after completing the steam and waterjet cleaning, the fan shall be operated at 5075 rpm for 15 minutes to verify that the impeller rotates smoothly. The fan shall be examined for evidence of damage or deformation. A sample of lubricant shall be taken from the bearings and tested to confirm that there has been no ingestion of water.

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4.7.8.4 Shock. To determine conformance to 3.4.8.5, the fan shall be mounted by normal mounting provisions to a shock machine capable of producing the shock pulse shown in figure 2. Three sawtooth shocks of 40 g and time duration of 11 ms (see figure 2) in each direction shall be applied along three mutually perpendicular axis of the fan (total of 18 shocks). After the shock test, the fan shall be operated at 5,075 rpm for 15 minutes to verify that the impeller rotates smoothly and examined for evidence of damage or deformation.

4.7.8.5 Vibration. To determine conformance to 3.4.8.6, the fan shall be attached by its normal mounting means either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. The fan shall be vibrated along each axis in accordance with the test level specified in figure 1. The sweep time shall be 15 minutes for a sweep frequency range of 5 to 500 to 5 Hz. When the fan resonance, below 5 Hz are measured, the test curves shall be extended to 2 Hz and the sweep time shall be 18 minutes. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 3. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep time shown on figure 3 for the specified range. The sinusodial cycling time shall be 30 minutes per axis. After the vibration test, the fan shall be subjected to the tests specified in 4.7.5 and 4.7.6. The fan shall be examined for evidence of damage or deformation.

4.7.9 Durability. When specified (see 6.2), durability testing shall be performed by the Government.

4.7.9.1 Fan system compatibility verification. To determine conformance to 3.4.6.1, two preproduction fans shall be installed in two test vehicles. The test vehicles shall be operated under conditions that approximate vehicle initial production tests, for a minimum of 6,000 miles each. The road conditions and mileage apportionment shall be as specified in 3.4.6.1. Fan system mating compatibility and performance under vehicle operational environment shall be evaluated for conformance to 3.4.6.1. Durability failures shall be observed and recorded.

4.7.9.2 Durability verification. To determine conformance to 3.4.6.2, the fan durability shall be verified by experiencing no durability failures during 6,000 miles of vehicles operation. A durability failure, as defined in 3.4.6.2, is the inability of the fan to perform its required function during testing and subsequently satisfy the performance tests of 4.7.4 through 4.7.7. Fans that must be replaced due to damage incurred as a result of failure of another component of the test vehicle shall not be classified as a durability failure.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

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6. NOTES

6.1 Intended use. Fans covered by this specification are intended for use in the M2/M3 Bradley Fighting Vehicle Systems (FVS).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article inspection is not required (see 3.1).
c. If responsibility for inspection shall be other than as specified (see 4.1).
d. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
e. If inspection conditions shall be other than as specified (see 4.3).
f. Preproduction inspection, if required (see 4.4.1).
g. If preproduction sample size shall be other than as specified (see 4.4.1).
h. If durability testing shall be performed (see 4.4.1 and 4.7.9).
i. If initial production inspection is not required (see 4.4.2).
j. If initial production sample size shall be other than as specified (see 4.4.2).
k. If first article samples may be delivered on the contract (see 4.4.4).
l. If control test sampling shall be other than as specified (see 4.6).
m. Selection of applicable level and packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. “Recovered materials” means materials that have been collected or recovered from solid waste (see 6.3.1.1).

6.3.1.1 Solid waste. “Solid waste” means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

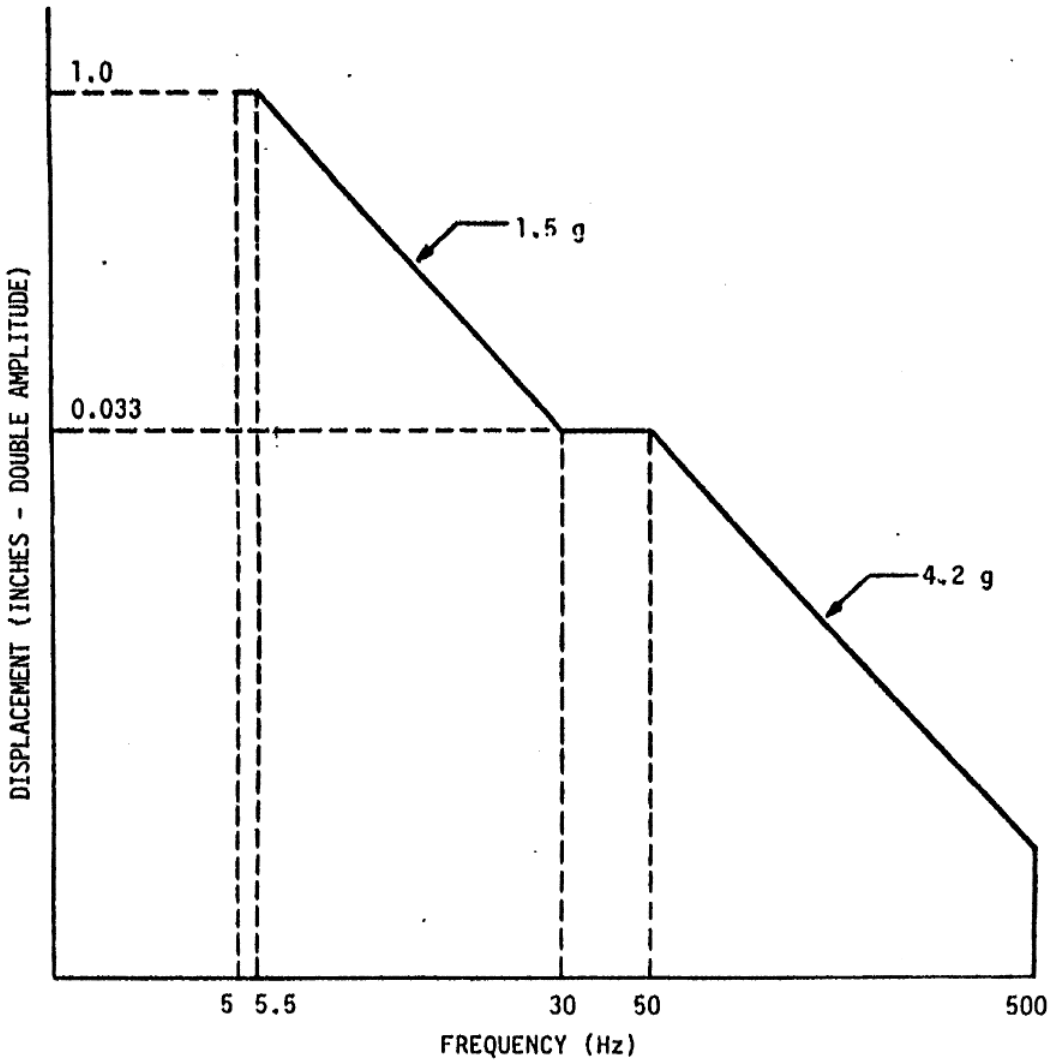
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6.4 Subject term (key word) listing.

Axial fan, vane
Engine coolant, fan, vaneaxial
Vaneaxial fan

6.5 Supersession data. This specification supersedes FMC specification 12292157, revision G, dated 27 October 1986; Fan, Vane Axial, Specification for.

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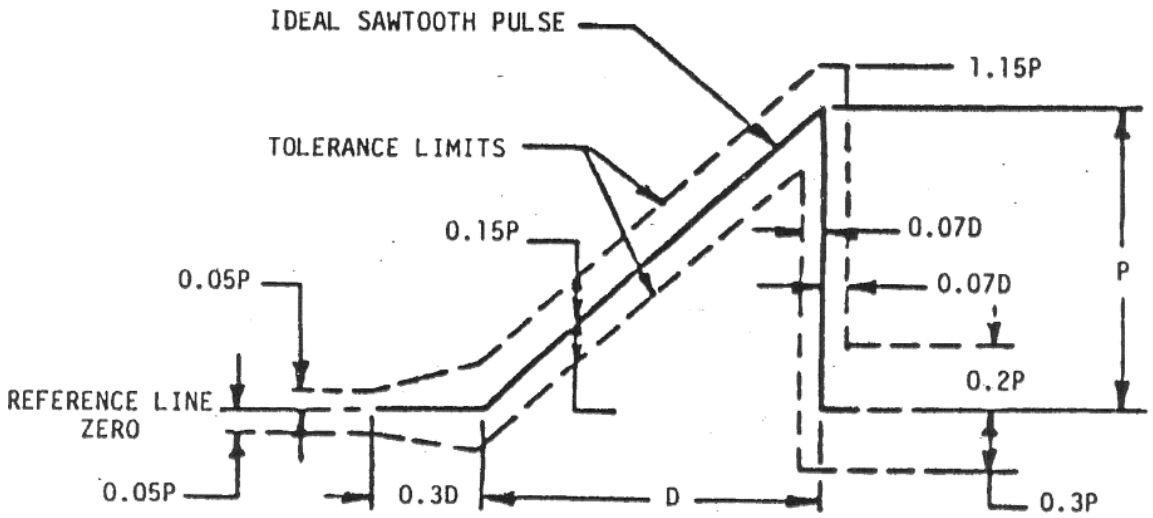


NOTES:

1. Acceleration levels: + g (peak).
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 1. Vibration test curve.

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P = Peak value = 40 g
D = Nominal duration = 11 ms

NOTE: The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and it's duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of $V_i \pm 0.1\ V_i$ where V_i is the velocity-change associated with the ideal pulse which equals 0.5 PD. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.lD after the pulse.

FIGURE 2. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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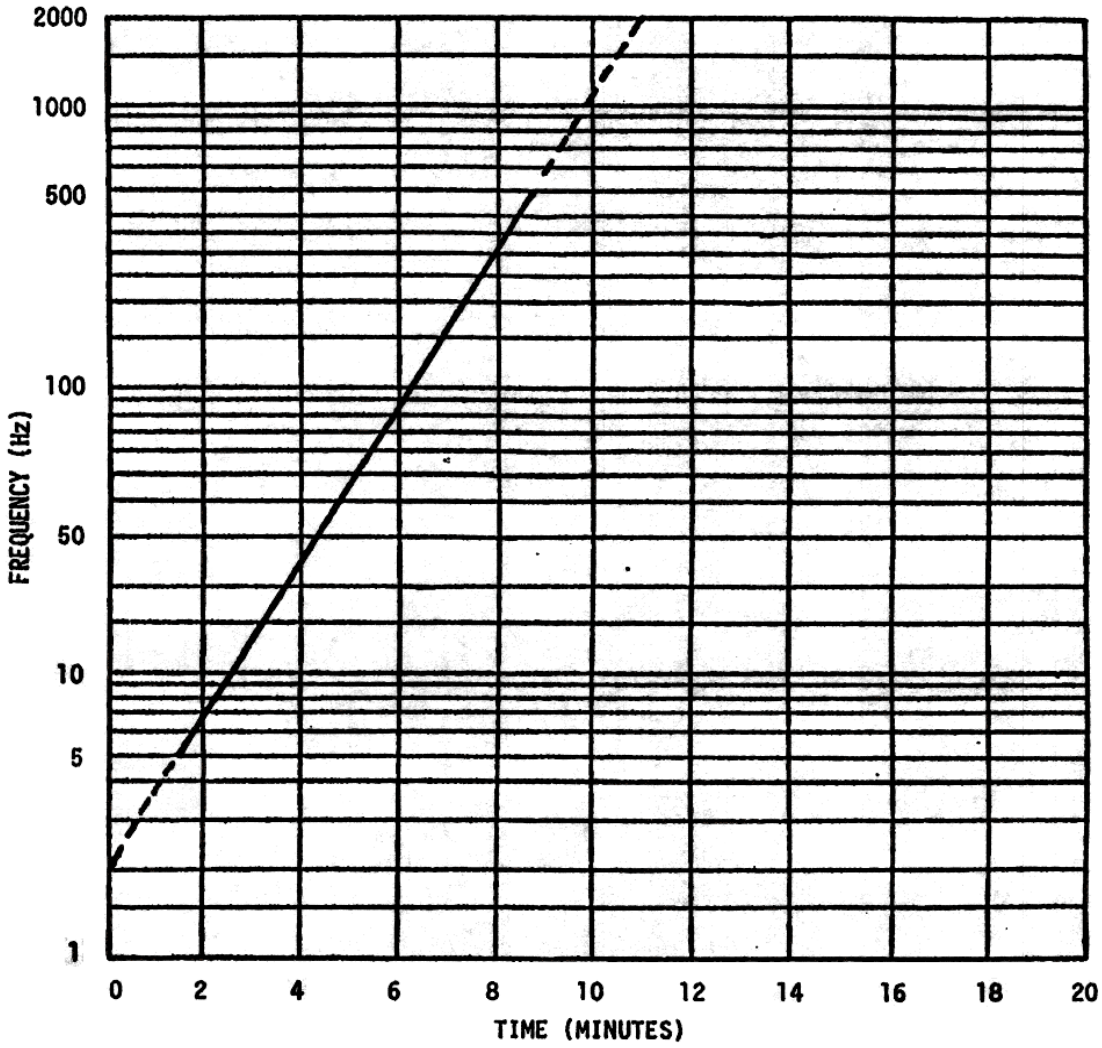


FIGURE 3. Logarithmic sweep.

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